

TOKMAK, Kyrgyzstan

WMO: 369110

Lat: 42.833N

Lon: 75.283E

Elev: 817

StdP: 91.89

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-13.9	-20.2	0.7	-15.8	-17.5	0.9	-12.7	10.3	8.2	8.1	6.4	0.6	90	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	34.8	19.9	33.3	19.3	31.9	18.8	21.2	32.0	20.3	30.9	19.6	29.9	1.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	14.0	26.0	16.7	13.2	25.0	15.8	12.4	24.6	65.9	32.0	62.4	31.0	59.6	30.0	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.4	5.2	4.1	DB	-20.7	37.5	3.9	1.1	-23.5	38.3	-25.8	39.0	-28.0	39.6	-30.8	40.4
(5)				WB	-21.0	23.2	3.8	1.2	-23.7	24.1	-25.9	24.7	-28.1	25.4	-30.8	26.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-1.7	0.1	7.2	12.9	17.8	22.3	24.5	23.5	18.1	11.9	5.3	-0.3	(6)
(7)		DBStd	10.34	5.78	5.70	5.65	4.93	3.68	3.28	2.91	3.22	4.04	4.83	5.51	5.78	(7)
(8)		HDD10.0	1311	364	279	122	27	1	0	0	1	36	159	322		(8)
(9)		HDD18.3	2942	622	511	346	171	54	7	1	3	53	204	392	579	(9)
(10)		CDD10.0	1988	1	2	35	114	242	369	449	418	244	95	18	1	(10)
(11)		CDD18.3	579	0	0	1	8	37	126	192	163	47	5	1	0	(11)
(12)		CDH23.3	6449	0	0	11	139	440	1346	2130	1756	542	79	6	0	(12)
(13)		CDH26.7	2689	0	0	1	37	122	544	992	792	184	17	1	0	(13)
(14)	Wind	WSAvg	1.3	1.3	1.2	1.5	1.4	1.3	1.3	1.2	1.1	1.1	1.1	1.4	(14)	
(15)	Precipitation	PrecAvg	452	26	30	52	72	73	37	19	11	17	42	43	27	(15)
(16)		PrecMax	751	70	81	121	138	155	92	74	46	50	93	97	101	(16)
(17)		PrecMin	296	6	4	17	11	25	5	0	0	14	11	1		(17)
(18)		PrecStd	92	13	17	22	33	34	20	17	9	13	21	23	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	17.5	24.9	30.5	32.2	35.9	37.3	36.1	33.2	28.8	23.6	15.3	(19)
(20)		0.4%	MCWB	5.7	7.5	11.7	16.0	17.7	19.4	20.8	20.7	18.3	14.7	11.9	6.8	(20)
(21)		2%	DB	10.8	13.5	22.1	27.1	29.7	33.4	35.3	34.4	30.9	25.4	19.1	12.2	(21)
(22)		2%	MCWB	4.2	6.1	10.4	14.6	17.1	19.1	20.3	19.9	17.5	13.7	10.1	5.1	(22)
(23)		5%	DB	8.4	10.6	19.3	24.4	27.8	31.6	33.4	32.7	28.8	22.7	16.1	9.9	(23)
(24)		5%	MCWB	2.8	4.7	9.6	13.2	16.5	18.6	19.6	19.2	16.6	12.4	8.6	4.0	(24)
(25)		10%	DB	6.2	8.1	16.3	21.8	25.8	29.9	31.8	31.1	26.5	20.1	13.0	7.7	(25)
(26)		10%	MCWB	1.8	3.3	8.7	12.5	15.8	18.1	19.2	18.7	15.5	11.5	7.2	3.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	8.7	13.1	17.8	19.6	21.6	22.7	22.5	19.7	15.9	13.1	7.6	(27)
(28)		0.4%	MCDWB	12.9	15.4	21.6	27.9	28.2	31.3	33.4	32.9	30.9	26.4	21.4	13.4	(28)
(29)		2%	WB	4.6	6.5	11.5	15.8	18.3	20.4	21.5	20.9	18.1	14.2	10.8	5.8	(29)
(30)		2%	MCDWB	9.7	12.4	19.7	23.9	26.9	30.0	32.4	32.1	29.1	23.5	17.9	11.0	(30)
(31)		5%	WB	3.3	5.1	10.3	14.4	17.4	19.5	20.5	20.0	17.0	13.2	9.1	4.5	(31)
(32)		5%	MCDWB	7.7	9.8	17.5	21.9	25.8	29.1	31.3	30.9	27.5	20.9	14.8	9.1	(32)
(33)		10%	WB	2.0	3.7	9.1	13.3	16.5	18.7	19.7	19.2	15.9	12.0	7.7	3.2	(33)
(34)		10%	MCDWB	5.6	7.7	15.5	20.1	24.5	28.1	30.3	29.7	25.6	18.9	12.3	7.4	(34)
(35)	Mean Daily Temperature Range	MDBR	10.1	10.1	11.0	11.7	12.5	12.7	12.8	13.4	13.8	13.0	10.6	9.9	(35)	
(36)		5% DB	MCDBR	12.4	13.6	15.4	16.1	15.3	14.8	14.5	14.7	16.0	16.8	14.9	12.4	(36)
(37)		5% DB	MCWBR	7.3	7.5	7.1	7.1	6.9	6.6	6.5	6.7	7.4	7.8	7.6	7.2	(37)
(38)		5% WB	MCDWR	11.3	12.1	13.7	13.9	13.7	13.4	13.7	13.7	15.0	14.7	13.6	11.4	(38)
(39)		5% WB	MCWBR	6.9	7.0	7.0	6.8	6.8	6.5	6.7	6.8	7.4	7.4	7.3	6.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.362	0.427	0.459	0.436	0.440	0.433	0.403	0.386	0.398	0.363	0.313	(40)	
(41)		taud	2.179	2.085	1.987	1.970	2.120	2.156	2.195	2.268	2.255	2.107	2.131	2.197	(41)	
(42)		Ebn at Noon	793	812	800	808	841	836	838	853	838	760	724	760	(42)	
(43)		Edh at Noon	96	126	158	173	153	149	141	127	119	122	101	86	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.15	3.07	4.00	4.95	6.10	6.57	6.64	6.24	5.02	3.24	2.08	1.75	(44)	
(45)		RadStd	0.15	0.23	0.30	0.41	0.46	0.33	0.40	0.33	0.25	0.33	0.20	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.48	-0.88	N/A	N/A	N/A	N/A

Nomenclature: See separate page